Understanding
Microsoft Fabric

Traditional **Challenge**

Multiple tools

Multiple vendors

Data Integration

Data Science

Data Engineering

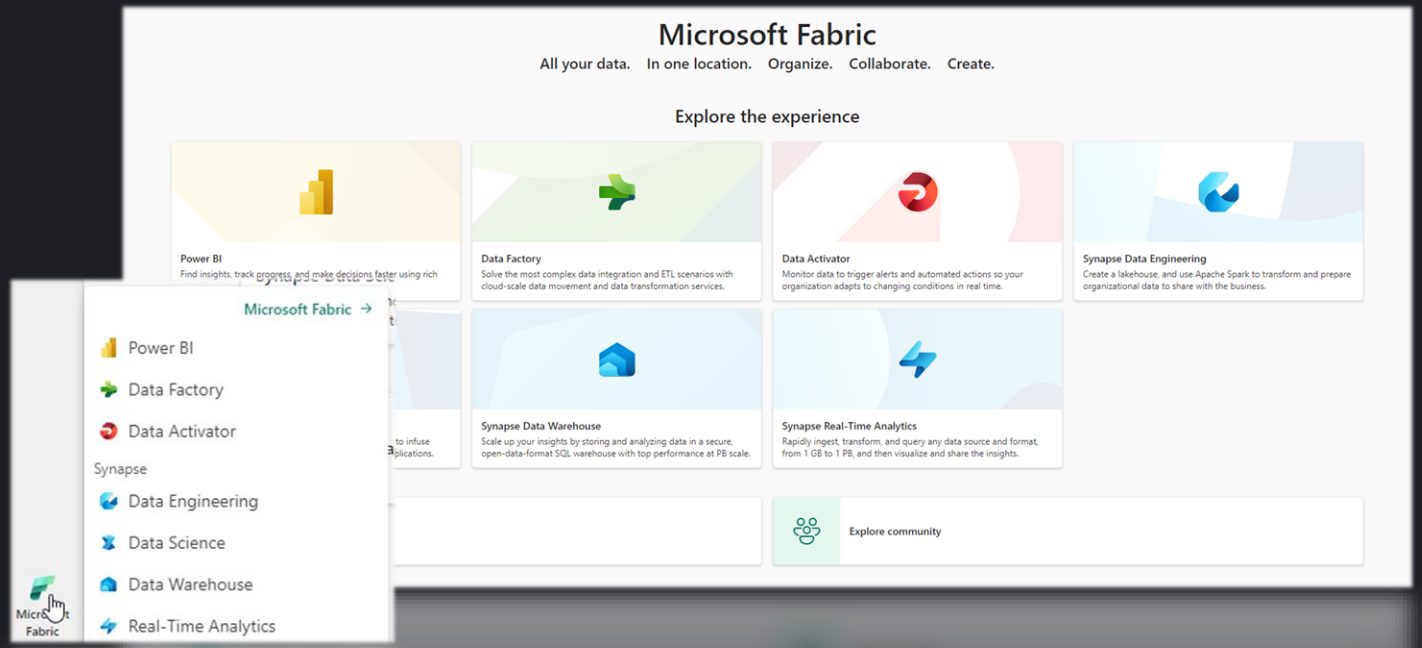
All-in-One analytics solution

SaaS

Data Factory | Data Science | Data Warehousing | Data Engineering | Power BI | Real-Time Analytics

OneLake

All-in-One analytics solution



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All-in-One analytics solution

Data Factory | Data Science | Data Warehousing | Data Engineering | Power BI | Real-Time Analytics

OneLake

OneLake & Lakehouses

Data Factory | Data Science | Data Warehousing | Data Engineering | Power BI | Real-Time Analytics

OneLake

Central data storage for all data

Shortcuts to external files

OneLake & Lakehouses

Data Factory | Data Science | Data Warehousing | Data Engineering | Power BI | Real-Time Analytics

Workspace A

- Contain all items
- Reports, Lakehouses etc.

Workspace B

OneLake & Lakehouses

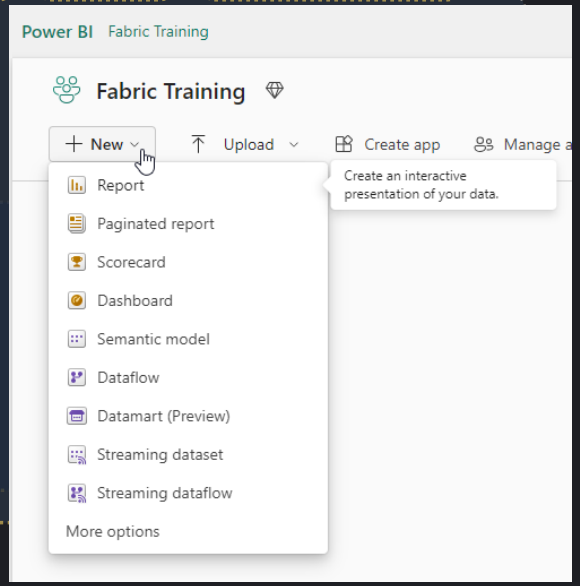
Data Factory | Data Science | Data Warehousing | Data Engineering | Power BI | Real-Time Analytics

Workspace A

- Contain all items
- Reports, Lakehouses etc.

Workspace B

- Setup users and permissions
- Objects can be shared across workspeaces



Lakehouses

Data Factory | Data Science | Data Warehousing | Data Engineering | Power BI | Real-Time Analytics

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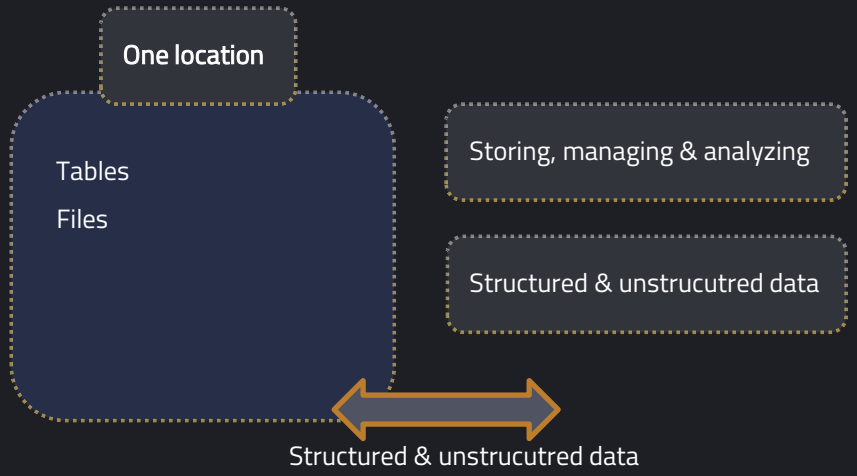
Lakehouses



- Storing and analyzing
- Structured and unstructured data in a single location.

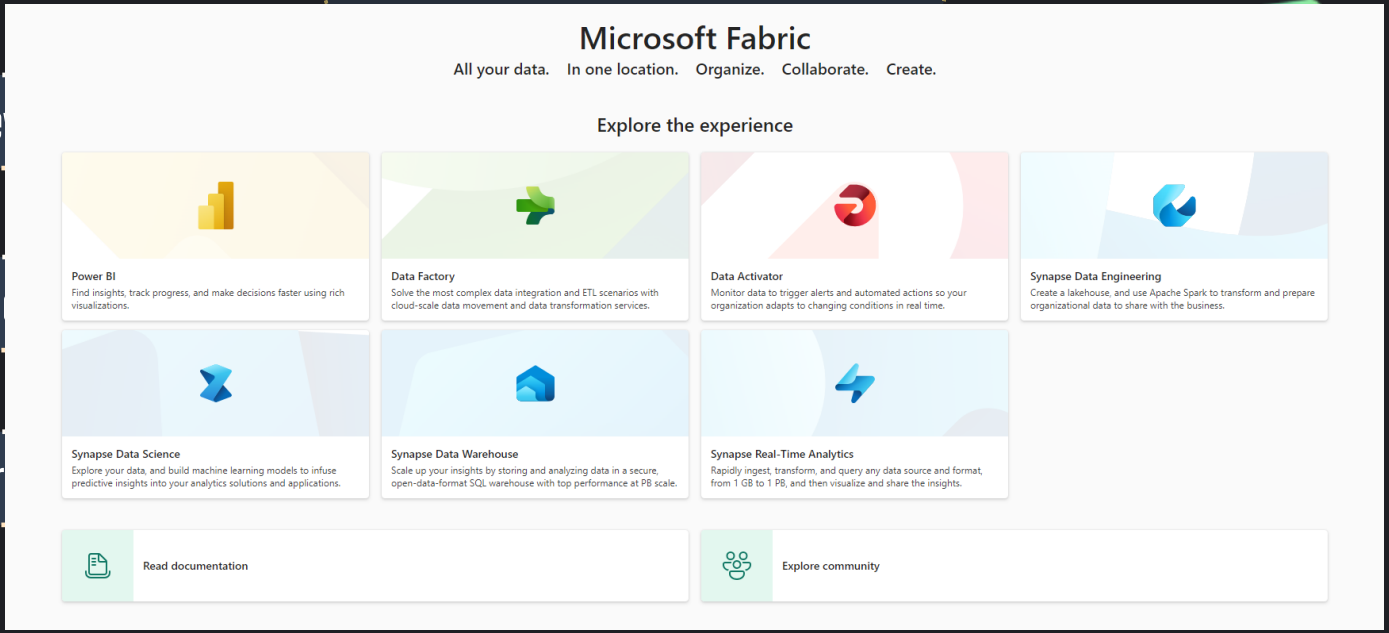
Create Microsoft account

- ✓ signup.live.com
- ✓ Create Microsoft account

Microsoft 365 De

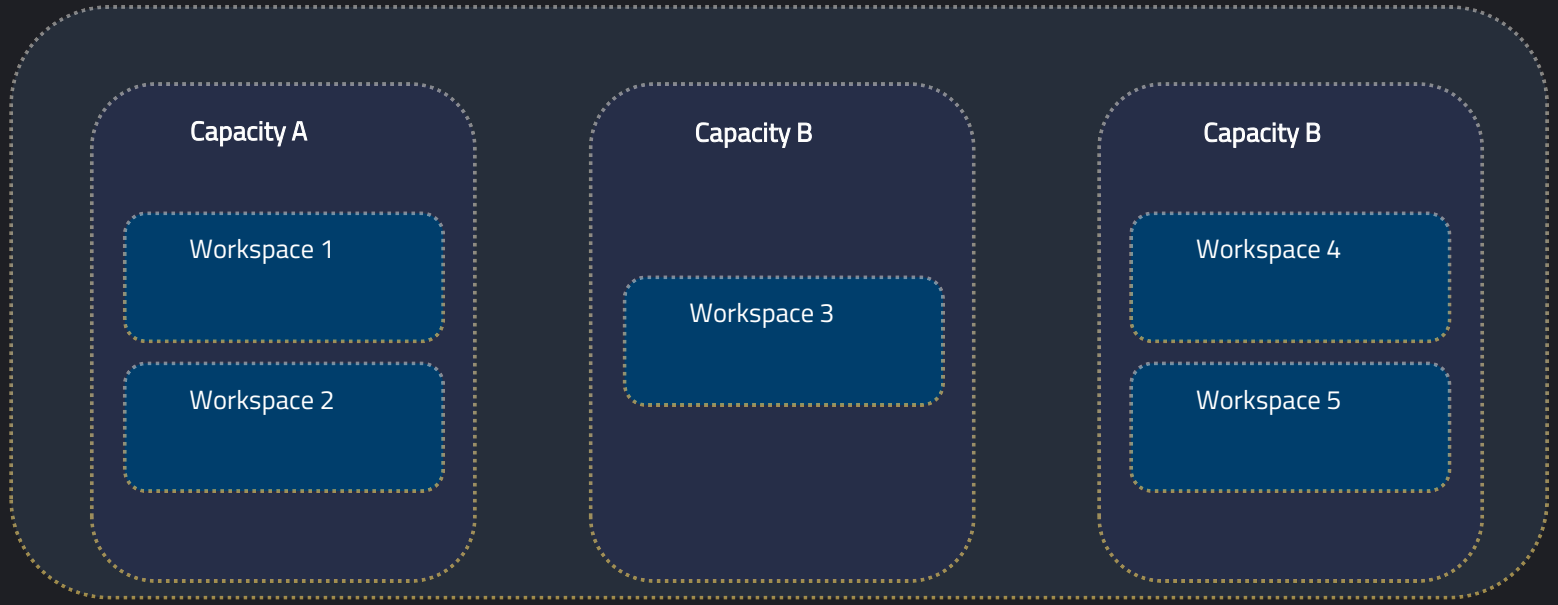
Fabric a

Fabric Fr



Structure

Tenant



Structure

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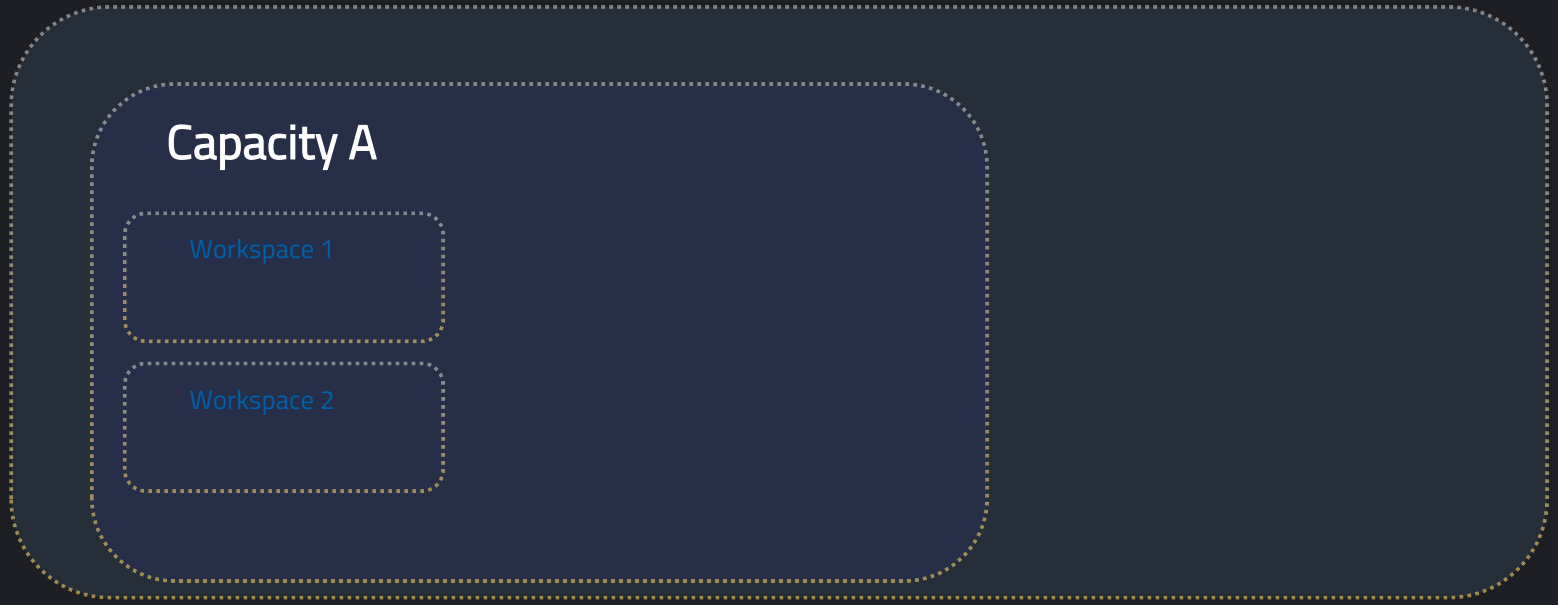

Spark

Introduction

- Apache Spark commonly used in big data processing and analytics workflows.

Strengths of Apache Spark

- Known for super-fast processing
- Distributed computing framework
- Utilizes in-memory processing, reducing the need to read from slow disk storage.
- Supports multiple programming languages: Scala, Python, and Java.
- Extends beyond batch processing to handle real-time data streaming, machine learning, etc.

How Apache Spark Works

- Utilizes a cluster of machines (nodes) for distributed data and processing tasks.
- Efficiently processes large datasets by splitting them into smaller, manageable partitions.
- Parallel processing across a cluster significantly speeds up data processing.

Resilient Distributed Datasets (RDDs)

- Core concept in Spark, serving as building blocks for data operations.
- Fault-tolerant, able to recover from errors or node failures by tracking data lineage.

Advantages of RDDs

- Ensures data integrity and reliability through fault tolerance.
- Optimizes data processing speed by storing as much as possible in memory.
- Cluster expansion is easy by adding more machines without significant code changes.

Deployment Options for Apache Spark

Complex Setup

- Acknowledges potential complexity and configuration challenges.
- May create overhead, especially for those new to distributed computing.

Fabric Deployment

- Zero set-up effort
- Supported languages
  - ✓ PySpark (Python)
  - ✓ Spark (Scala)
  - ✓ Spark SQL
  - ✓ SparkR